



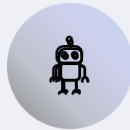
What to Expect From AI in 2026

Personal Agents, Mega Alliances, and the Gigawatt Ceiling

Executive summary of Goldman Sachs' analysis on the transformative forces reshaping the artificial intelligence landscape and their profound implications for business, policy, and infrastructure development.

Three Converging Forces Reshaping the AI Landscape

The AI industry is undergoing a fundamental transformation that moves beyond standalone software to integrated, autonomous systems. Three interconnected forces are driving this shift, each with cascading effects across the technology ecosystem.



Agent-as-a-Service Revolution

The industry is transitioning from selling software licenses to delivering autonomous work execution.

Specialized multi-agent systems now act as personal digital coworkers, executing complex, multi-step tasks across enterprise platforms without human intervention.



The Gigawatt Ceiling

Physical infrastructure, not algorithmic intelligence, has become the primary constraint on AI development. The computational power required for massive models is hitting hard limits of local power grids, forcing a fundamental rethinking of scaling strategies.



Mega Alliances

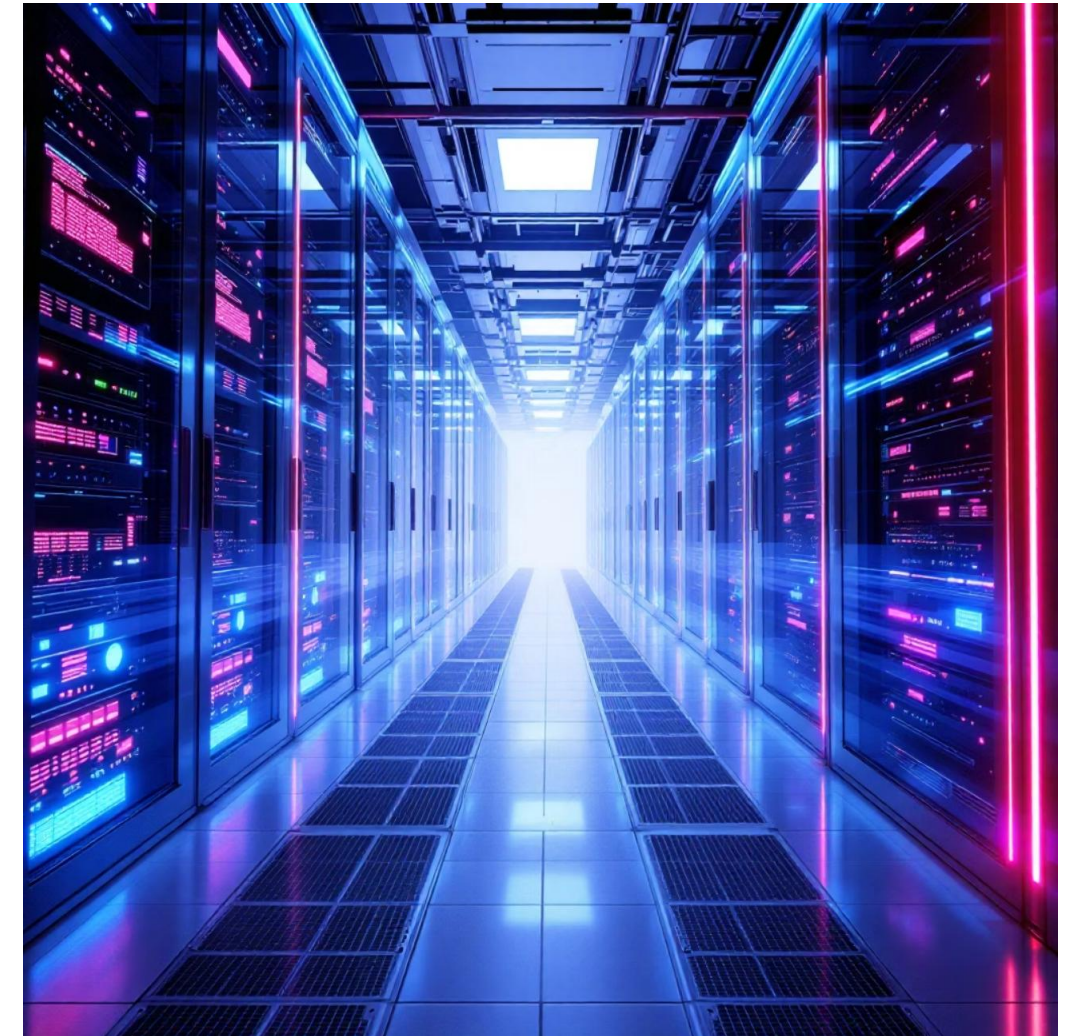
Exorbitant capital requirements are driving unprecedented consolidation. Deep partnerships between chip manufacturers, cloud providers, and energy companies are forming to secure the resources needed for continued AI expansion.

The Energy Bottleneck: When Physics Meets Computing

The report presents a sobering reality: AI development is now constrained by the laws of physics rather than algorithmic innovation. Training and running massive AI models requires computational power that's straining local power grids to their absolute limits.

Energy availability measured in gigawatts has become the new hard ceiling for tech growth. This fundamental constraint is forcing a strategic pivot away from simply building larger baseline models toward more efficient "agentic" workflows that maximize output per watt.

The implications extend far beyond technical considerations. Data centers are increasingly becoming matters of national security and energy policy, requiring governments and utilities to grapple with resource prioritization between public infrastructure and corporate AI development.





The End of the Hourly Economy: Token-Based Work

As fleets of AI agents autonomously execute routine tasks, the traditional model of compensating human labor based on hours worked faces existential disruption. The report predicts a macro-economic transformation where work is valued and billed based on outputs and compute tokens consumed rather than human time spent.



Traditional Model

Compensation based on hours worked and billable time



Token Economy

Value measured by outputs delivered and computational resources consumed

The New Premium Skill: Orchestration Over Execution



The definition of highly skilled work is fundamentally changing. Where technical execution once dominated value creation, the premium skill is now managing, prompting, and orchestrating teams of digital agents.

The human role transforms to defining intent, setting boundaries, and reviewing outputs. Workers who excel at articulating requirements, decomposing problems, and guiding autonomous systems will command disproportionate value in this new economy.

This shift has profound implications for education, training, and talent development across all sectors, requiring a fundamental rethinking of what skills organizations should cultivate.



Environmental and Policy Challenges Ahead

Grid Prioritization Dilemmas

Local utilities face impossible choices between powering homes, businesses, and AI data centers. Should residential customers lose power so tech companies can train larger models? These questions move from hypothetical to urgent.

Carbon Impact Accountability

The massive power demands of AI translate directly to carbon emissions. As climate commitments intensify, corporations and governments must address the environmental cost of computational expansion.

National Security Concerns

Control over energy infrastructure and computing resources becomes a matter of strategic national interest. The same data centers powering AI also concentrate economic and military advantage.

The Datacenter Construction Boom

The "Gigawatt Ceiling" translates directly into an unprecedented pipeline of capital projects for the Architecture, Engineering, and Construction industry. Tech companies are frantically building next-generation data centers and retrofitting local power grids, creating massive demand for specialized AEC expertise.

3.2x

Construction
Growth

Projected increase in
data center construction
spending through 2026

\$56B

Market Value

Estimated annual
investment in new data
infrastructure and grid
upgrades

85%

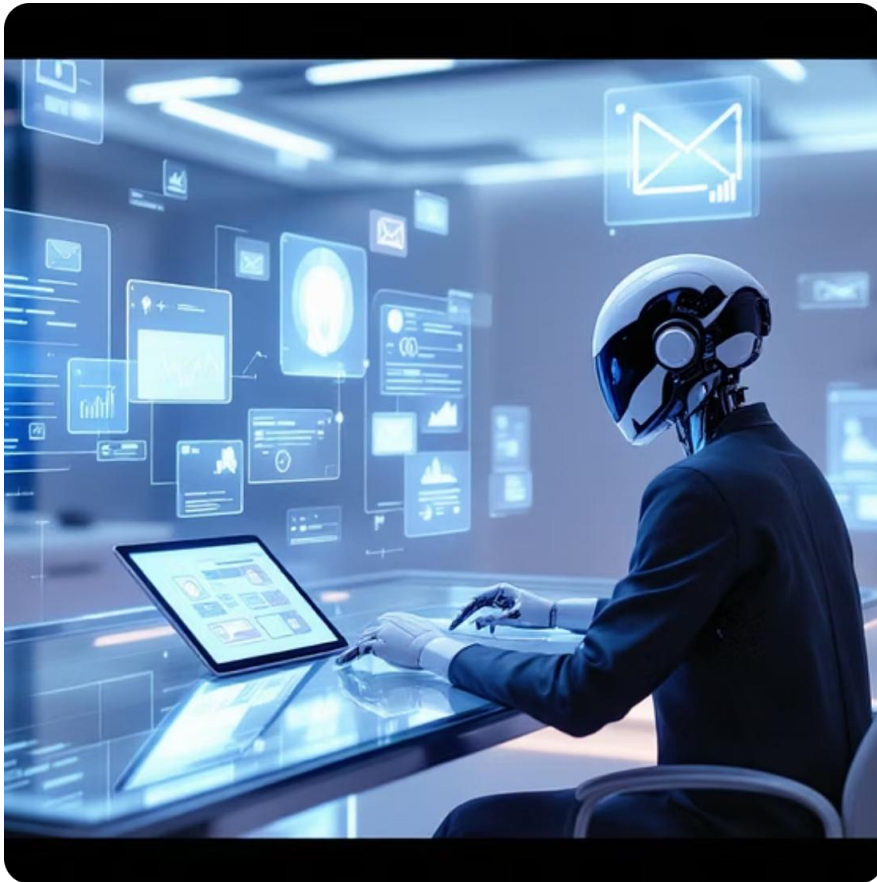
Specialized
Requirements

Portion of projects
requiring advanced
structural engineering
for high-density server
loads



Engineering Applications: From Manual to Autonomous

The "agent-as-a-service" model will fundamentally alter day-to-day design routines for AEC professionals. Rather than manual setup and verification, engineers will deploy autonomous agents to handle routine execution while focusing on judgment and oversight.



Intelligent Briefing Agents

Autonomous systems parse unread emails and calendars to compile comprehensive project briefings before morning meetings, synthesizing information from multiple sources.



Code Compliance Verification

Agents automatically verify baseline parameters against ACI 318, ASCE 7, and other standards, pulling guidelines from internal repositories and cross-referencing initial designs before human review.



Specialized Infrastructure Design

Advanced liquid cooling systems and integrated energy substations require specialized designs to support massive loads of high-density server racks and power distribution.

Business Model Transformation for AEC Firms

Current Model: Billable Hours

- Revenue tied to time spent on projects
- Manual execution as primary value driver
- Pricing based on estimated hours
- Profitability constrained by labor costs

Future Model: Value-Based Pricing

- Revenue tied to outcomes delivered
- AI efficiency as competitive advantage
- Pricing based on project value
- Profitability amplified by automation

When AI agents calculate shear wall loads, detail masonry, or size sewer lines in minutes rather than hours, traditional billable hour models collapse. Firms must overhaul pricing strategies, shifting toward fixed-fee or value-based approaches that capitalize on internal AI systems' speed and efficiency.

The competitive advantage moves from having the most engineers to having the most sophisticated agent orchestration.

Strategic Imperatives for AEC Leaders

01

Invest in Data Center Expertise

Build specialized capabilities in high-density structural loads, advanced cooling systems, and integrated power infrastructure to capture construction boom opportunities.

03

Restructure Business Models

Transition from hourly billing to value-based pricing that reflects the efficiency gains from AI systems, repositioning firms as outcome-delivery partners.

02

Develop Agent Orchestration Skills

Train teams to manage autonomous systems effectively, focusing on problem decomposition, requirement articulation, and quality oversight rather than manual execution.

04

Engage Energy Policy

Participate in grid planning and energy policy discussions to ensure infrastructure development supports both AI expansion and public utility needs.

The AI revolution is not coming—it's already here. AEC firms that move decisively to embrace these changes will position themselves as indispensable partners in the infrastructure buildout required for the next generation of artificial intelligence.